



Ride Fare Surge Analysis – Insight Summary

Project Objective

The objective of this project was to analyze ride booking data to understand customer behavior, booking trends, revenue patterns, surge pricing, and operational performance. The analysis was performed using Python for data preprocessing and exploratory data analysis (EDA), followed by the development of an interactive Power BI dashboard for business reporting.

Dataset Summary

- Total ride bookings were analyzed to evaluate booking performance and customer behavior.
- The dataset contains ride status, booking value, vehicle type, customer and driver ratings, pickup and drop locations, payment methods, ride distance, and cancellation details.
- Data cleaning and feature engineering were performed before visualization.

Key Business Insights

1. Booking Performance

- Completed rides accounted for the majority of total bookings.
- Customer and driver cancellations represented a relatively smaller share, indicating stable service performance.

2. Revenue Analysis

- Revenue varied across different vehicle categories.
- Premium vehicle categories generated higher revenue per ride.
- Peak hours contributed significantly to overall revenue.

3. Surge Pricing

- Surge pricing occurred mainly during high-demand hours.
- The analysis showed a clear relationship between booking volume and surge pricing.

4. Customer Experience

- Average customer ratings remained above 4, indicating a generally positive user experience.
- Most customers preferred digital payment methods such as UPI.

5. Operational Insights

- Certain pickup and drop locations consistently recorded higher ride demand.
- Driver cancellation reasons highlighted operational challenges such as long pickup distances and availability issues.

Dashboard Overview

Page 1 – Executive Overview

Provides a high-level summary of overall business performance using KPIs, revenue analysis, booking status, payment methods, and vehicle-wise revenue.

Page 2 – Operations & Surge Analysis

Focuses on ride demand patterns, surge pricing, ride distance, fare analysis, booking trends, and operational performance.

Page 3 – Customer & Cancellation Insights

Analyzes customer ratings, driver ratings, cancellation trends, pickup/drop locations, and cancellation reasons to understand customer experience.

Recommendations

- Improve driver availability during peak hours to reduce surge pricing.
- Optimize driver allocation for high-demand pickup locations.
- Reduce pickup distance to minimize driver cancellations.
- Continue promoting digital payment methods due to high customer adoption.
- Monitor vehicle-wise revenue trends to improve fleet utilization.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook
- Power BI
- DAX
- Git & GitHub

Conclusion

This project demonstrates an end-to-end data analytics workflow, beginning with raw data preprocessing and exploratory analysis in Python, followed by the creation of an interactive Power BI dashboard. The findings provide meaningful business insights into ride booking performance, revenue generation, customer behavior, surge pricing, and operational efficiency.